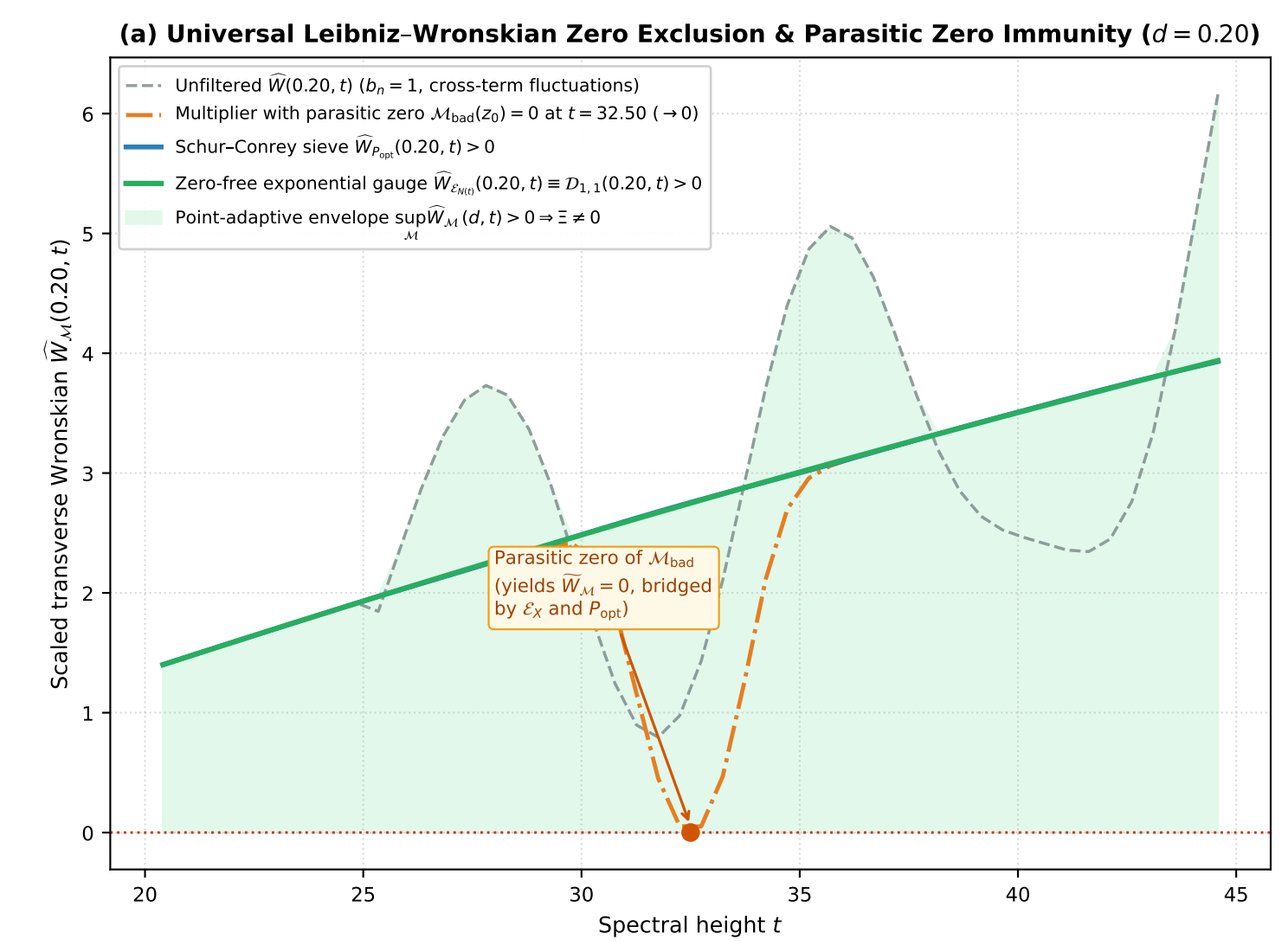
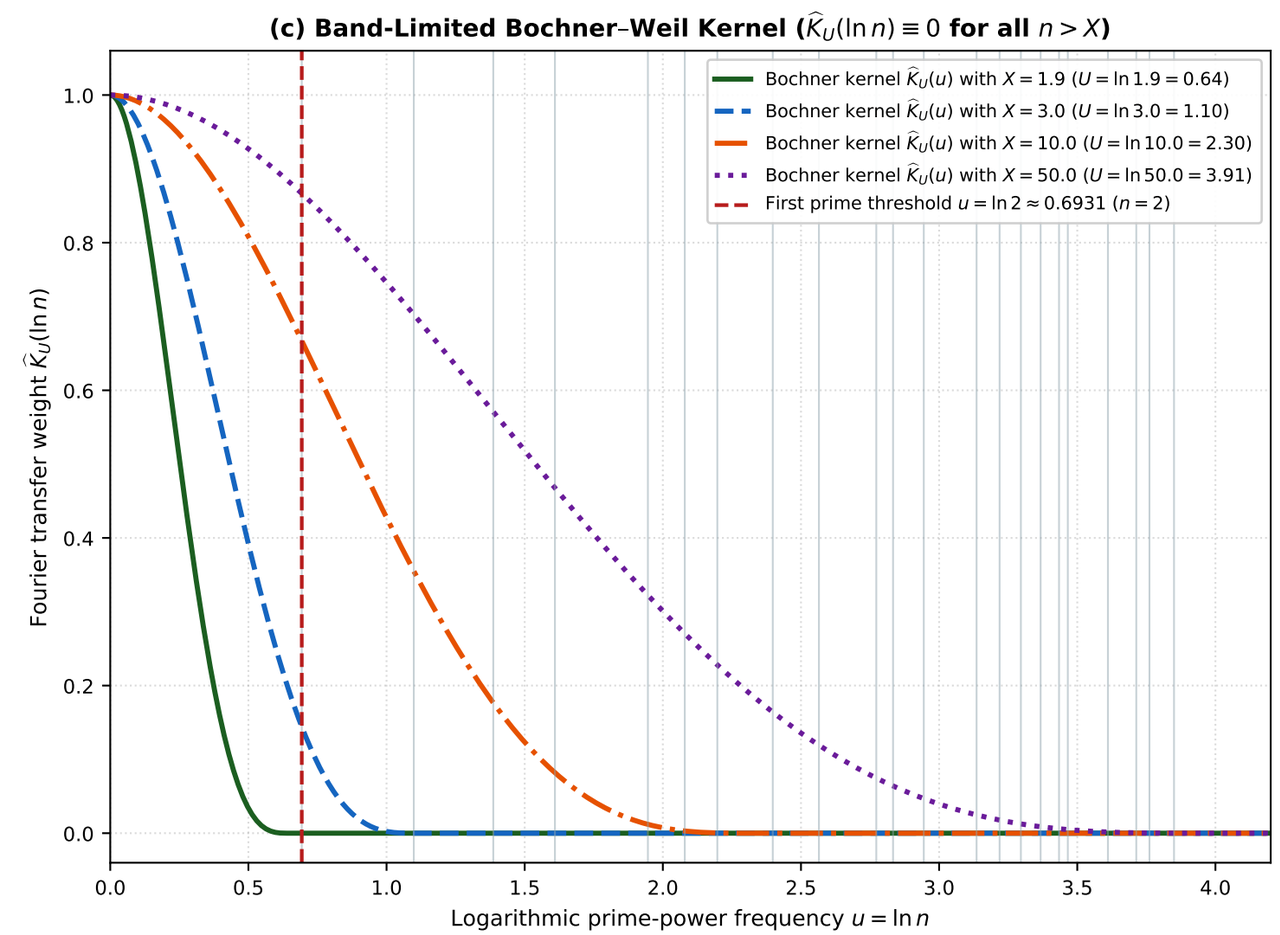

(a) Universal Leibniz–Wronskian Zero Exclusion & Parasitic Zero Immunity ($d = 0.20$)
Unfiltered $\widehat{W}(0.20, t)$ ($b_n = 1$, cross-term fluctuations)
Multiplier with parasitic zero $\mathcal{M}_{\text{bad}}(z_0) = 0$ at $t = 32.50$ ($\to 0$)
Schur–Conrey sieve $\widehat{W}_{P_{\text{opt}}}(0.20, t) > 0$
Zero-free exponential gauge $\widehat{W}_{\mathcal{E}_{N(t)}}(0.20, t) \equiv \mathcal{D}_{1,1}(0.20, t) > 0$
Point-adaptive envelope $\sup_{\mathcal{M}} \widehat{W}_{\mathcal{M}}(d, t) > 0 \Rightarrow \Xi \neq 0$
Parasitic zero of $\mathcal{M}_{\text{bad}}$ (yields $\widehat{W}_{\mathcal{M}} = 0$, bridged by $\mathcal{E}_X$ and P_{opt})
Scaled transverse Wronskian $\widehat{W}_{\mathcal{M}}(0.20, t)$
Spectral height t
(b) Sieved Saddle-Point Wronskian Minus Gabcke–Olver Bound on $d = 0$
Unfiltered $\widehat{W}^{(0)}_{\text{main}} - B^{(0)}_{\widehat{W}}$ (dips < 0 at Lehmer pairs)
Unfiltered $\widehat{W}^{(C_0)}_{\text{main}} - B^{(C_0)}_{\widehat{W}}$ (-0.0934 at $t = 221.16$)
Schur–Conrey $P_{\text{opt}}(x)$: $\widehat{W}_{P_{\text{opt}},\text{main}} - B^{(C_0)}_{\widehat{W}} > 0$ ($+12.35$ at dip)
Exponential gauge $b^{\mathcal{E}}_n(N(t))$: $\widehat{W}_{\mathcal{E},\text{main}} - B^{(C_0)}_{\widehat{W}} > 0$ ($+12.64$ at dip)
Uncertified region (≤ 0)
Sieve $X = 10$: $b^{\mathcal{E}}_{2..10} \equiv 0$ (0% overshoot)
$b^{\mathcal{E}}_n(10) \in [0, 1]$
$|b^{P_{\text{opt}}}_n(10)|$
Certified lower bound $\widehat{W}_{X,\text{main}}(0, t) - B_{\widehat{W}}(0, t)$
Spectral height t


(c) Band-Limited Bochner–Weil Kernel ($\widehat{K}_U(\ln n) \equiv 0$ for all $n > X$)
Bochner kernel $\widehat{K}_U(u)$ with $X = 1.9$ ($U = \ln 1.9 = 0.64$)
Bochner kernel $\widehat{K}_U(u)$ with $X = 3.0$ ($U = \ln 3.0 = 1.10$)
Bochner kernel $\widehat{K}_U(u)$ with $X = 10.0$ ($U = \ln 10.0 = 2.30$)
Bochner kernel $\widehat{K}_U(u)$ with $X = 50.0$ ($U = \ln 50.0 = 3.91$)
First prime threshold $u = \ln 2 \approx 0.6931$ ($n = 2$)
Fourier transfer weight $\widehat{K}_U(\ln n)$
Logarithmic prime-power frequency $u = \ln n$
(d) 2D Continuum Certification of $\widehat{W}_{X,\text{main}}(d, t) - B^{(C_0)}_{\widehat{W}}(d, t) > 0$
100.0000% Continuum Certified
(1,002,000 / 1,002,000 cells on $[0, 0.50] \times [14, 1000]$)
$\min(\widehat{W}_{X,\text{main}} - B^{(C_0)}_{\widehat{W}}) = +0.5696 > 0$
Transverse offset $d = \sigma - 1/2 \in [0, 0.50]$
Spectral height $t \in [14, 1000]$
$\log_{10}(\widehat{W}_{P_{\text{opt}},\text{main}}(d, t) - B^{(C_0)}_{\widehat{W}}(d, t))$